What Drives Firms to Go Green?

Sophiko Skhirtladze | Tue/Thurs Section 4 | GitHub: sskhirtladze

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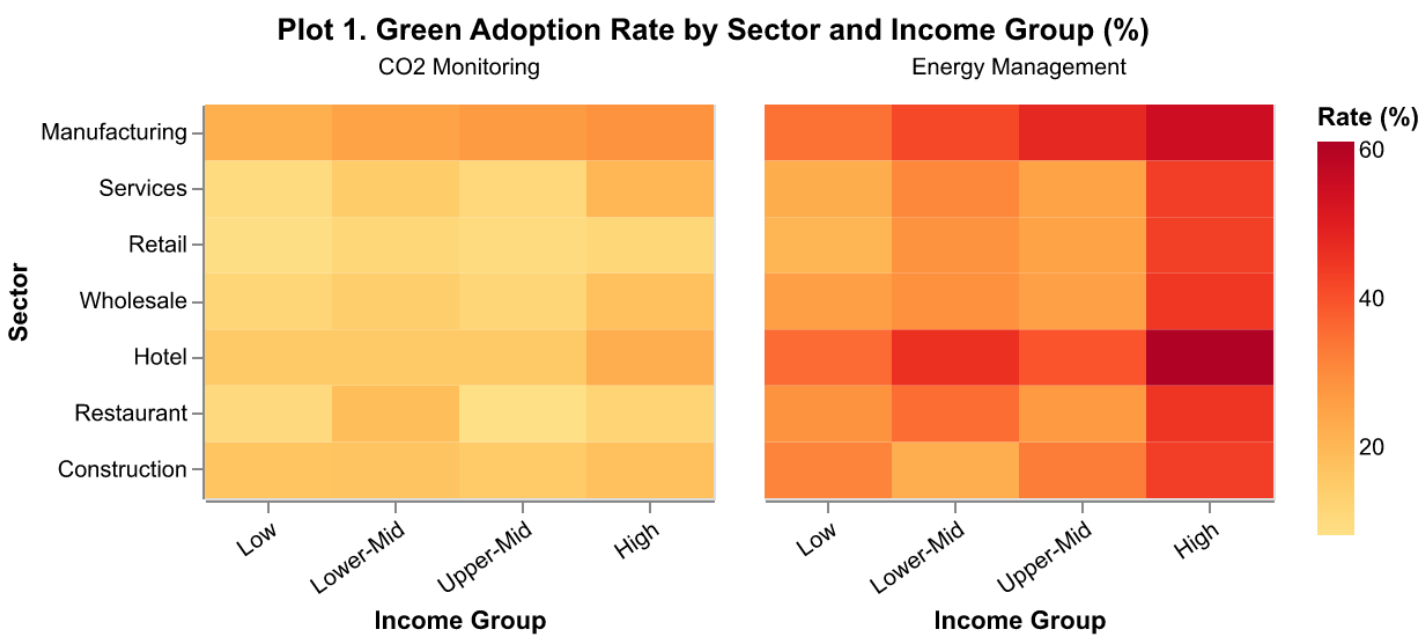
Research Question

This paper examines the determinants of firms' green practices using the data from the World Bank Enterprise Surveys, focusing on two specific behaviors: whether firms monitor their CO_2 emissions, and whether they adopt energy management measures to reduce emissions, waste, or pollution. Both are drawn from the Green Economy module of the World Bank Enterprise Surveys and capture concrete, operational steps firms take in response to environmental pressures. The key question is whether institutional quality and environmental governance, captured by the Environmental Performance Index and B-Ready Environment Index, significantly influence these adoption decisions beyond what can be explained by firm characteristics and country economic and political context. The three visualizations below explore adoption patterns by sector and income group, the marginal effects of key predictors from the controls model, and the relationship between national EPI scores and country fixed effects extracted from the logit models.

Approach & Data

The pipeline has four stages. First, country-level data are assembled by merging WDI, WGI, EPI, and B-Ready sources, with the B-Ready environmental index constructed manually by identifying environmentally tagged columns and summing them into a composite score. Second, the Enterprise Survey data are cleaned, including recoding negative survey values to missing, harmonizing variable names across countries that received different questionnaire versions, and merging in the country-level dataset. Third, two sets of logit models are estimated: fixed effects models with country and year dummies, and controls models that replace country dummies with macro and governance covariates.

Adoption Rate by Sector and Income Group

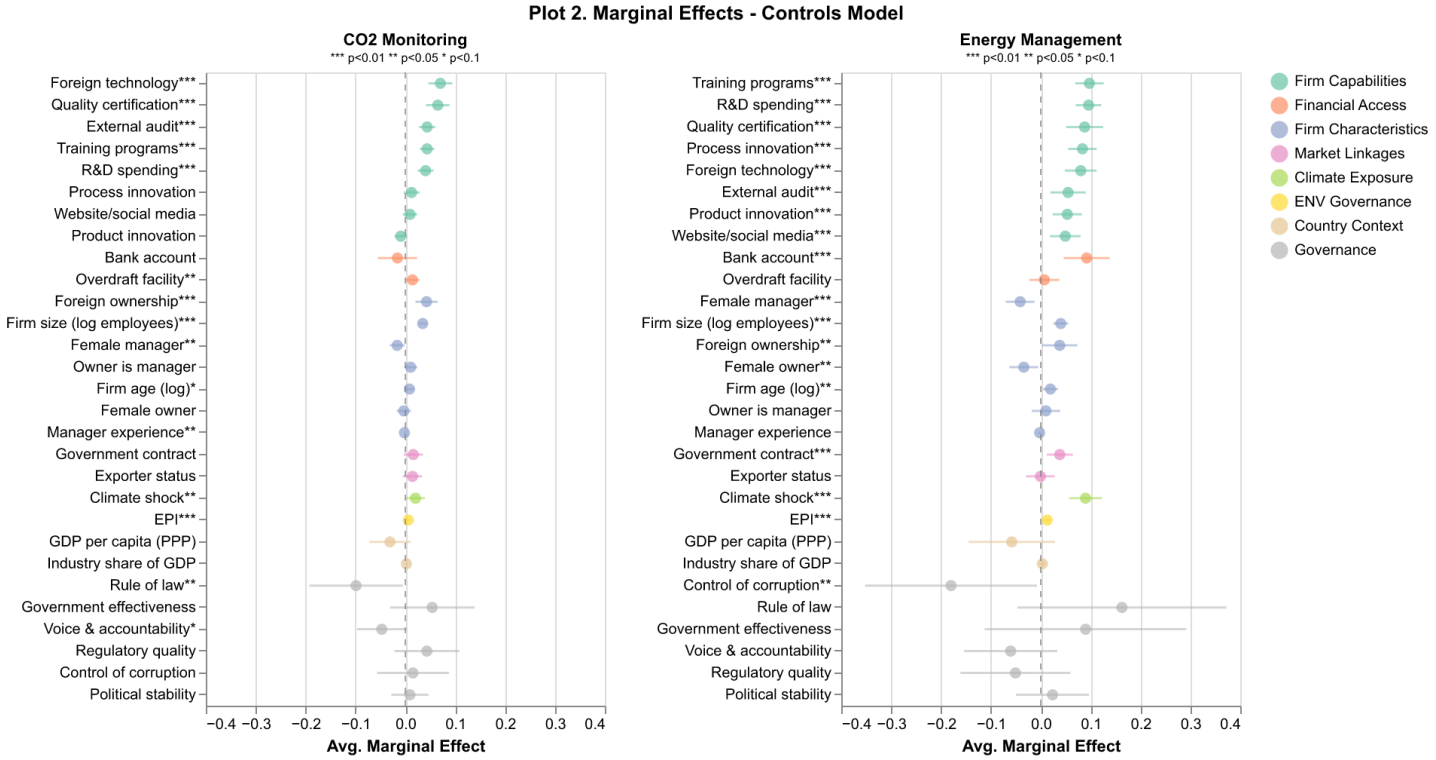


Plot 1 shows green adoption rates by sector and income group as a faceted heatmap. Higher-income countries display markedly higher adoption across all sectors, with manufacturing leading. CO_2 monitoring stays below 20% even in upper-middle-income groups, while energy management adoption reaches around 40% in high-income groups. Services and retail lag behind manufacturing, especially for CO_2 monitoring. This underscores the uneven diffusion of green practices, with clear income and sectoral gradients. The next question is what drives these patterns once we control for firm and country characteristics.

Marginal Effects with Firm and Country Controls Model

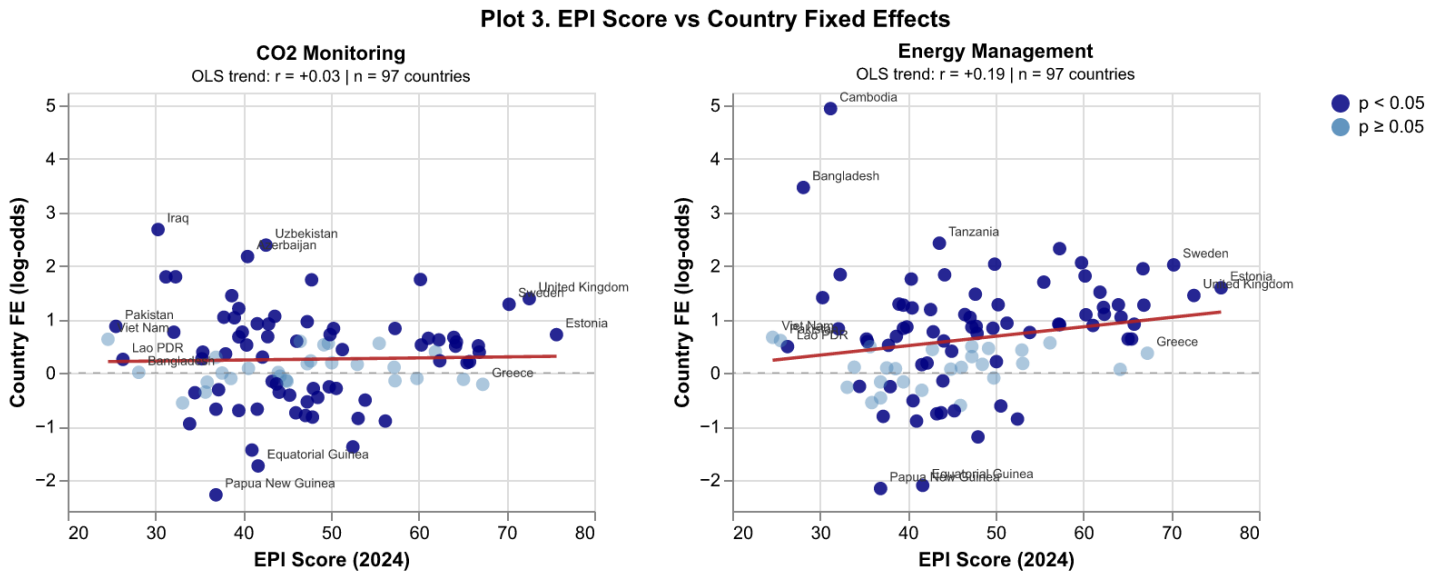
Plots 2 displays average marginal effects from the controls model for CO_2 monitoring and energy management respectively. Firm capability variables, including foreign technology use, quality certification, training programs, and R&D spending, consistently show the largest positive and statistically significant effects across both outcomes, confirming that organizational capacity and knowledge access are the primary drivers of green adoption. Financial access variables are largely insignificant, suggesting credit constraints are not the binding barrier. Experience with climate shocks and government procurement contracts also raises the probability of adoption across both outcomes, pointing to the role of realized physical risk and public sector incentives as indirect drivers of environmental behavior.

At the country level, EPI enters positively and significantly for both outcomes, indicating that firms operating in countries with stronger national environmental performance are more likely to adopt green practices even after controlling for firm characteristics and economic context. By contrast, the B-Ready environmental index is neither economically nor statistically significant, suggesting that the environmental content of business regulations as currently measured does not translate into observable firm behavior. Governance indicators are mixed, largely imprecise, and sometimes counterintuitive: rule of law enters negatively for CO_2 monitoring and control of corruption negatively for energy management, likely reflecting multicollinearity among governance dimensions and the difference between mandatory and voluntary adoption across institutional contexts. Taken together, the country-level results speak directly to the research question: national environmental performance matters, but general institutional quality has only a marginal and inconsistent influence on green adoption.



EPI Score vs Country Fixed Effects

Plot 3 shows country fixed effects from the FE logit models against EPI scores. The near-flat trend for CO_2 monitoring (r = +0.03) contrasts with the stronger positive relationship for energy management (r = +0.19), suggesting EPI captures institutional conditions that enable systematic energy practices more than emissions reporting behavior. Each point is a country, colored by whether the FE coefficient is statistically significant. The top outliers by residual and EPI score are labeled to highlight interesting cases that deviate from the overall trend, such as countries with high EPI but low FE or vice versa. This plot illustrates that even after controlling for firm characteristics there is significant heterogeneity in adoption of sustainable practices across countries only marginally explained by the EPI index.



Streamlit App

The interactive dashboard, deployed at https://sskhirtladze-final-project-data-viz-sophik-codedashboard-b5qxmv.streamlit.app/, lets users explore the country-level dimension of the research question interactively. The choropleth map visualizes adoption rates and EPI sub-indicators across all surveyed countries, while the country profile panel shows how any selected country compares to the cross-country average on green adoption, EPI sub-scores, and B-Ready environmental scores. This connects directly to the research question by making the cross-country variation in environmental governance and firm behavior visible and navigable in a way static plots cannot.